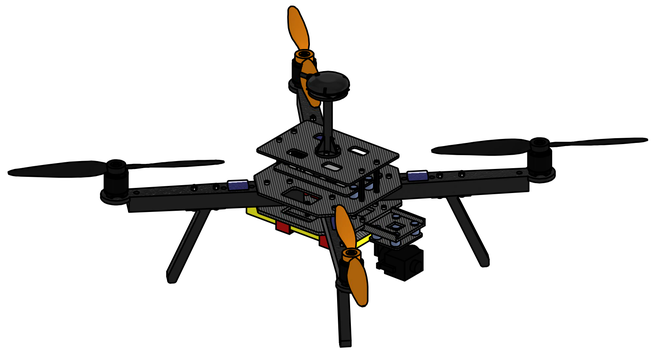
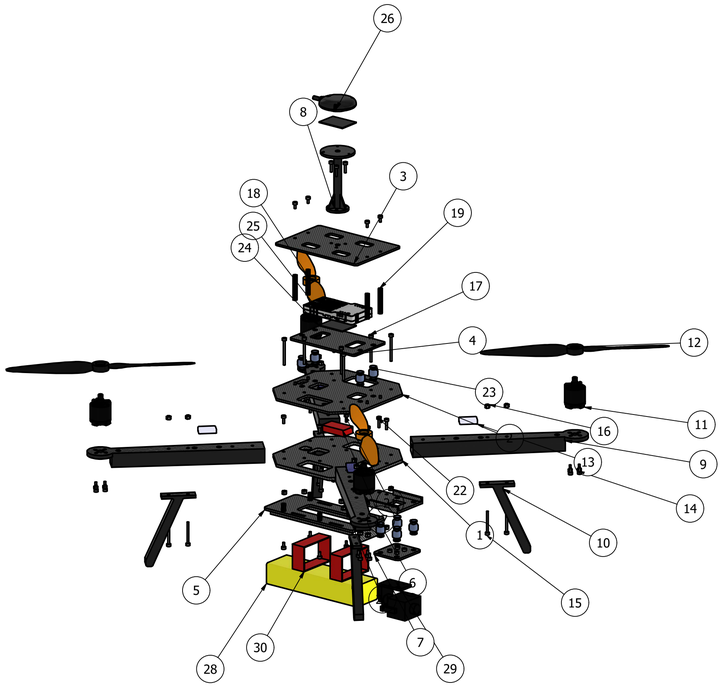
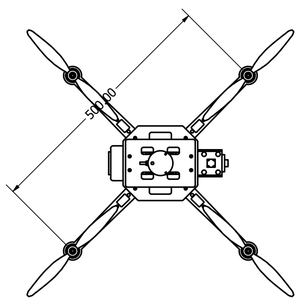






PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
16	23	ISO 10511 M3	Nylon-insert lock nut M3, A2 stainless
17	8	ISO 4762 M3x35	Socket head cap screw M3x35, A2 stainless
18	2	T1045-CCW	Propeller 10x4.5 in (1045) CCW, glass-fibre nylon, T-Motor T1045 class
19	4	STANDOFF-M3x35-FF	Hex standoff M3x35 female-female, 5 mm AF, 6061 aluminium
20	4	STANDOFF-M3x10-FF	Hex standoff M3x10 female-female, 5 mm AF, 6061 aluminium
21	4	ISO 4762 M3x8	Socket head cap screw M3x8, A2 stainless
22	7	ISO 4762 M3x12	Socket head cap screw M3x12, A2 stainless
23	8	DAMPER-BALL-12	Silicone anti-vibration ball damper, 12 mm ball, M3-plate type
24	2	FOAM-3M-4026	Double-sided mounting foam 40x30x1 mm (FC)
25	1	PIXHAWK-6C-ALU	Flight controller Holybro Pixhawk 6C (aluminium case) 84.8x44x12.4
26	1	HOLYBRO-M9N	GPS Holybro M9N (u-blox NEO-M9N + IST8310 compass), 50 mm puck
27	1	PM02-V3	Power module Holybro PM02 V3: voltage / current sensing + 5.2 V for the Pixhawk only (XT60 in / out, no ESC pads; the 4 ESCs are fed by the QD-603 PDB)
28	1	GA-B-45C-5000-4S1P	LiPo 4S1P 14.8 V 5000 mAh 45C (Gens Ace), 154x46x30, XT60
29	1	GIMBAL-2AX-MICRO	2-axis micro brushless gimbal with 1/2.3 in action camera (roll + pitch)
30	2	STRAP-20x250	Battery strap 20 mm hook-and-loop with rubberised grip (loop through tray slots)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	QD-101	Bottom centre plate 150x150x4, arm clamp + battery cage + gimbal mount
2	1	QD-102	Top centre plate 150x150x3, FC damper mounts + hatch standoffs
3	1	QD-105	Electronics bay hatch (top cover), 4 thumb-screw standoffs, GPS mast base
4	1	QD-106	FC damping plate on 4 silicone ball dampers
5	1	QD-107	Battery tray / retention cage: strap slots, side rails, front stop
6	1	QD-108	Upper gimbal bracket (frame side of the damper set), ribbed
7	1	QD-109	Lower gimbal damping plate (gimbal side)
8	1	QD-110	GPS mast, hollow (GPS cable inside), 3 gussets, M9N platform
9	4	QD-103	Arm: tapered ribless box printed solid-in-CAD with 4 walls + 20 % gyroid (respec 2026-10-29: the sealed hollow bays failed the print check), cable channel, integral motor pad with 16-19 mm slotted M3 holes
10	4	QD-104	Landing leg, tapered strut, bolts under the arm (2x M3x30)
11	4	AIR2216-II-KV920	Brushless motor T-Motor AIR2216 II KV920 (4S, 1045 prop), 16x19 mm M3 mount; buy 2 CW + 2 CCW shaft-thread versions to match the self-tightening T1045 props (no prop nut)
12	2	T1045-CW	Propeller 10x4.5 in (1045) CW, glass-fibre nylon, T-Motor T1045 class
13	4	XROTOR-MICRO-30A	ESC 30 A BLHeli_S 2-4S (Hobbywing XRotor Micro 30A class), 23.8x14.5x5.8
14	32	ISO 4762 M3x6	Socket head cap screw M3x6, A2 stainless
15	8	ISO 4762 M3x30	Socket head cap screw M3x30, A2 stainless

QUADDRONE - 500 CLASS GPS QUADCOPTER
CAD MASS 1694 g (INVENTOR), AUW 1774 g INCL. 80 g WIRING/RX ALLOWANCE
MOTORS T-MOTOR AIR2216 II KV920, T1045 PROPS, 4S 5000 mAh
TWR 3.00 (16.0 V) / 2.57 (14.8 V)
HOVER 14.7 MIN (80% DOD, 242 W)
FRAME: 3D PRINTED PA6-CF, PARAMETRIC (QuadDroneParameters.ipt)
EXPLODED VIEW = POSITIONAL REPRESENTATION 'Exploded' IN QuadDrone.iam
FASTENERS ISO 4762 / ISO 10511 A2, THREADS COSMETIC

DRAWN	Jay Patel	10/29/2026	
CHECKED			
QA			TITLE: QuadDrone - general assembly
MFG			
APPROVED			
SIZE	D	DWG NO	QD-DWG-001 REV
SCALE	0.29 : 1	SHEET 1 OF 1	